

FictiPay Customer-Churn Prediction

Model Performance Report · v7 pipeline

This report documents a customer-churn model for the bKash-presents-NSUCEC Datathon. Churn is scored as **AUC-ROC** on a predicted probability `CHURN_PROB` $\in [0, 1]$. Three base learners (LightGBM, CatBoost, a quantile-normalised neural network) are trained on a single shared 5-fold split, averaged over 5 seeds, and combined through a weight-optimised ensemble selected automatically on out-of-fold (OOF) score. The final model reaches **OOF AUC = 0.98535**, with **Precision@10% = 0.9173** and **Recall@10% = 0.7235**. A dedicated leakage audit found no exploitable data artefact (max probe AUC 0.5016), so the score reflects genuine behavioural signal.

Final OOF AUC	Precision@10%	Recall@10%	Churn rate	Features
0.98535	0.9173	0.7235	12.68%	194

1. Problem & data

The objective is to predict whether a FictiPay account churns, observed as a snapshot at 2024-03-31 22:59:57. Three raw sources are joined per account: a transaction ledger (`TRX_DATETIME`, `SRC_ACCOUNT`, `DST_ACCOUNT`, `TRX_TYPE`, `TRX_AMT`), day-end balances (`AVAILABLE_BALANCE` by date), and KYC profile (`ACCOUNT_TYPE`, `GENDER`, `REGION`, `ACCOUNT_OPEN_DATE`). Five transaction types are observed: `MERCHANTPAY`, `P2P`, `BILLPAY`, `CASHIN`, `CASHOUT`. Aggregations are computed in DuckDB with spill-to-disk to handle larger-than-RAM scans, then assembled into a 194-column feature matrix.

Quantity	Value
Train rows / Test rows	595,000 / 255,000
Engineered features	194
Churn rate (positive class)	0.1268 (scale_pos_weight 6.89)
Accounts with sent / received features	848,216 / 687,806
Library versions	duckdb 1.3.2 lgb 4.6.0 xgb 3.2.0 cb 1.2.10

2. Methodology

Shared-fold OOF. All models use one `StratifiedKFold(5, random_state=42)` split so their OOF columns align and the ensemble comparison is fair. **5-seed averaging** (`[44, 43, 45, 46, 47]`) is applied within each fold for the tree models (the NN uses 2 seeds for runtime). **Per-fold rank-normalisation** of OOF and test predictions removes cross-fold probability-scale mismatch — the v7 fix that makes the pooled OOF score and the ensemble weighting honest. Class imbalance is handled with `scale_pos_weight = 6.89`.

***Note on tuning.** Hyperparameters were set directly (values below) rather than grid-searched, so this pipeline contains no learning-rate/depth sweep curves. The tuning that the notebook does perform and that the charts in Section 4 document is (a) per-fold early stopping on the number of boosting rounds, and (b) a 5,000-sample random search over ensemble weights with automatic model selection on OOF AUC.*

Model	Key settings (from notebook code)
LightGBM	lr 0.02 · num_leaves 128 · min_child_samples 60 · feature_fraction 0.75 · bagging_fraction 0.85 · L1 0.5 · L2 2.0 · n_estimators 6000 (early-stop 200) · GPU
CatBoost	lr 0.03 · depth 8 · l2_leaf_reg 6 · iterations 6000 (od_wait 200) · native_categoricals · GPU
Neural Net	MLP 256-128-64 · BatchNorm + Dropout(0.3/0.2) · Adam 1e-3 · QuantileTransformer(normal) · early-stop on val_auc (patience 8)

Ensemble

per-fold rank-norm → equal / Dirichlet-optimised / logistic-stacker candidates →
auto-select best OOF

3. Performance — cross-validation results

Per-fold and pooled out-of-fold AUC-ROC for each base model on the shared 5-fold split.

Model	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4	Pooled OOF
LightGBM	0.98492	0.98528	0.98577	0.98535	0.98535	0.98534
CatBoost	0.98474	0.98518	0.98563	0.98520	0.98517	0.98518
Neural Net	0.98461	0.98499	0.98549	0.98504	0.98520	0.98506

Table 1 — Per-fold validation AUC by model.

3.1 Ensemble selection & final metrics

Ensemble strategy	OOF AUC-ROC
Optimized weights ← selected	0.98535
Best single (LightGBM)	0.98534
Equal blend	0.98532
Logistic stacker	0.98531

Table 2 — The weight-optimised blend wins by only +0.00002 over the best single model (weights: LightGBM 0.745, CatBoost 0.115, Neural Net 0.140), i.e. the ensemble is a marginal, honest gain.

Final-model metric	Value
Selected strategy	Optimized weights (lgb 0.745 / cb 0.115 / nn 0.140)
Selected OOF AUC-ROC	0.98535
Precision @ top 10% risk	0.9173
Recall @ top 10% risk	0.7235
Max leakage-probe AUC	0.5016 (no exploitable artefact)

Table 3 — Final model card. At a 10%-of-population intervention budget the model captures ~72% of churners at ~92% precision.

4. Tuning & model-selection charts

The two decisions actually tuned in this pipeline — which base model and which ensemble weighting — are shown below, alongside the cross-validation stability that justifies trusting the OOF score.

5-Fold Cross-Validation Stability by Model

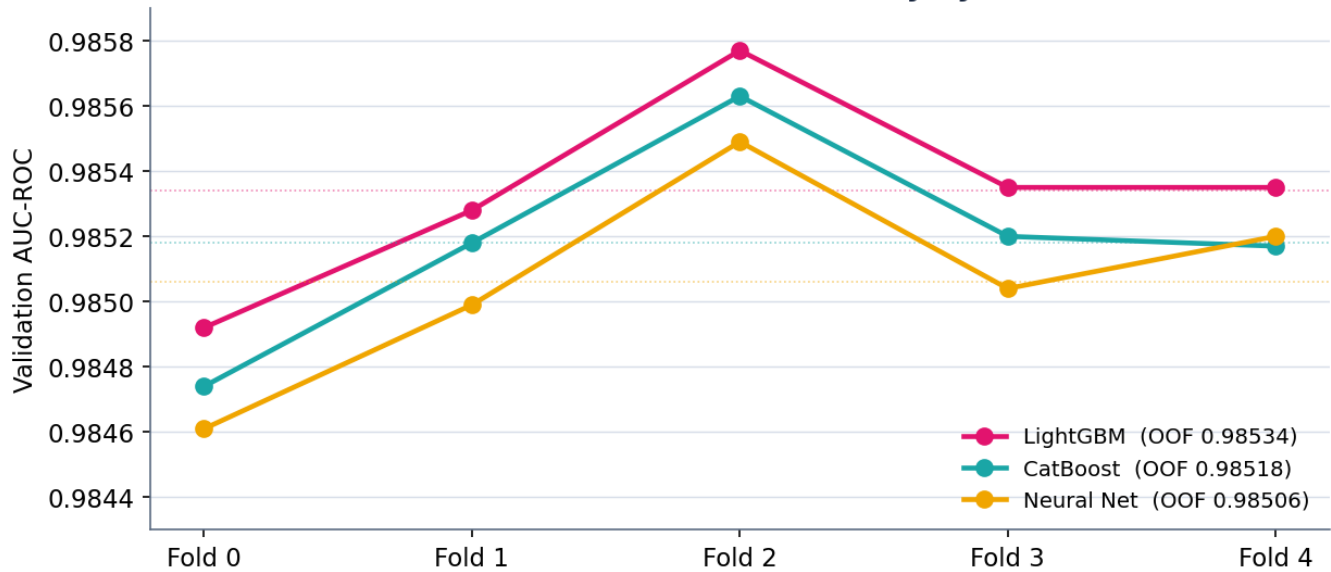


Figure 1 — Per-fold AUC for each model across the shared 5-fold split; dotted lines mark pooled OOF. Folds move together (Fold 2 easiest for all), confirming a stable split.

Ensemble Strategy Comparison (auto-selected on OOF)

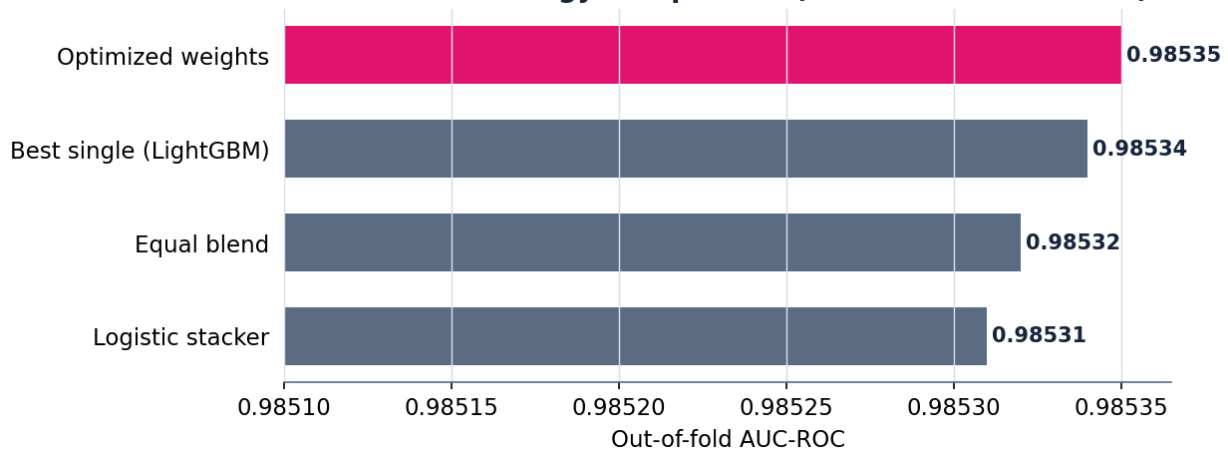


Figure 2 — Ensemble-strategy comparison on OOF AUC (the weight-tuning result). Optimised weights edge out the best single model; the spread across all four strategies is < 0.0001 .

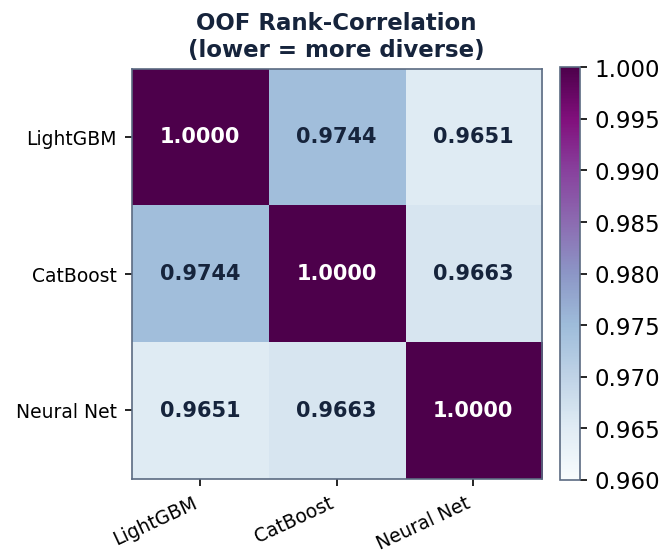


Figure 3 — OOF rank-correlation between models. The NN is the most diverse (0.965 vs LightGBM), which is why a small NN weight still helps the blend.

5. Feature diagnostics

Before modelling, each candidate feature was scored as a *single-feature* classifier (its own AUC against the label). Recency, recent-window counts, and the Poisson zero-probability dominate — several exceed 0.95 on their own.

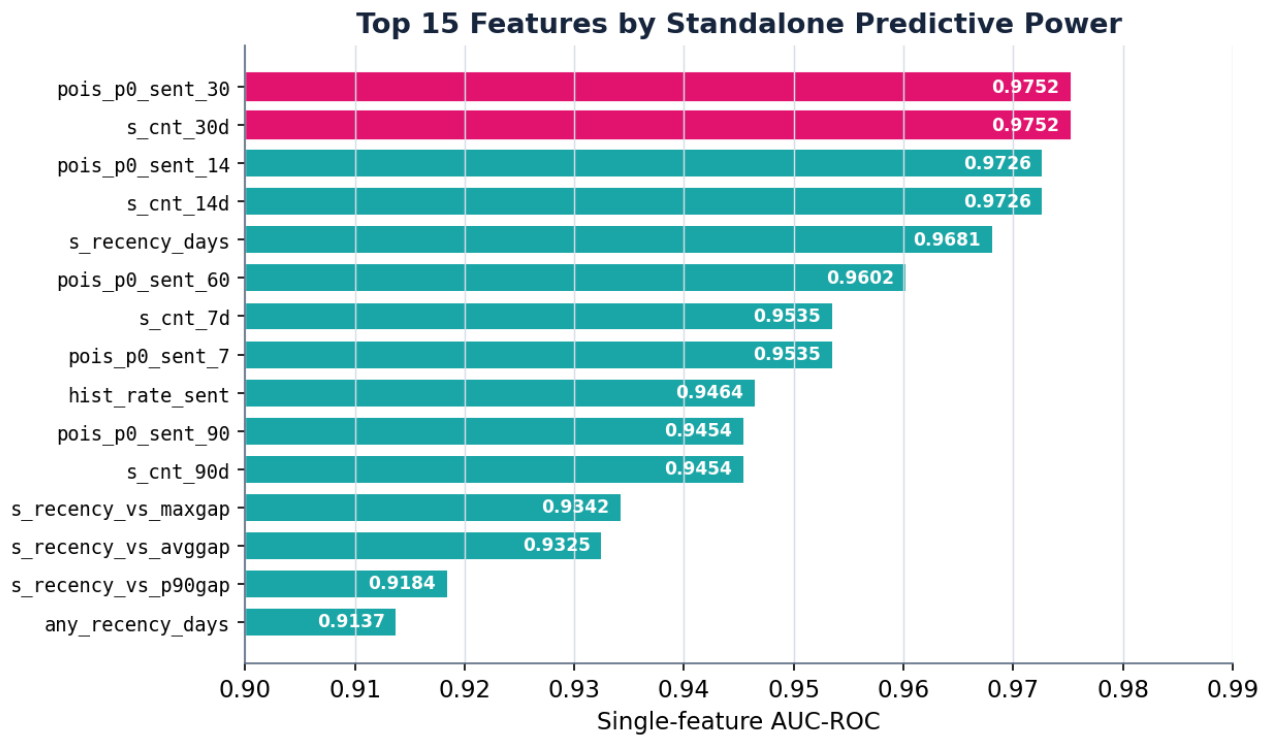


Figure 4 — Top 15 features by standalone AUC-ROC. The strongest single signal, *pois_p0_sent_30* / *s_cnt_30d*, reaches 0.9752.

6. Explainability (SHAP)

Global feature importance from the final LightGBM fold model, on a 20,000-row sample. The model leans on the 21-, 10-, and 7-day sent-transaction counts and on raw recency; balance-staleness (*days_since_bal_change*) and the periodicity feature (*days_to_next_expected*) contribute corroborating signal.

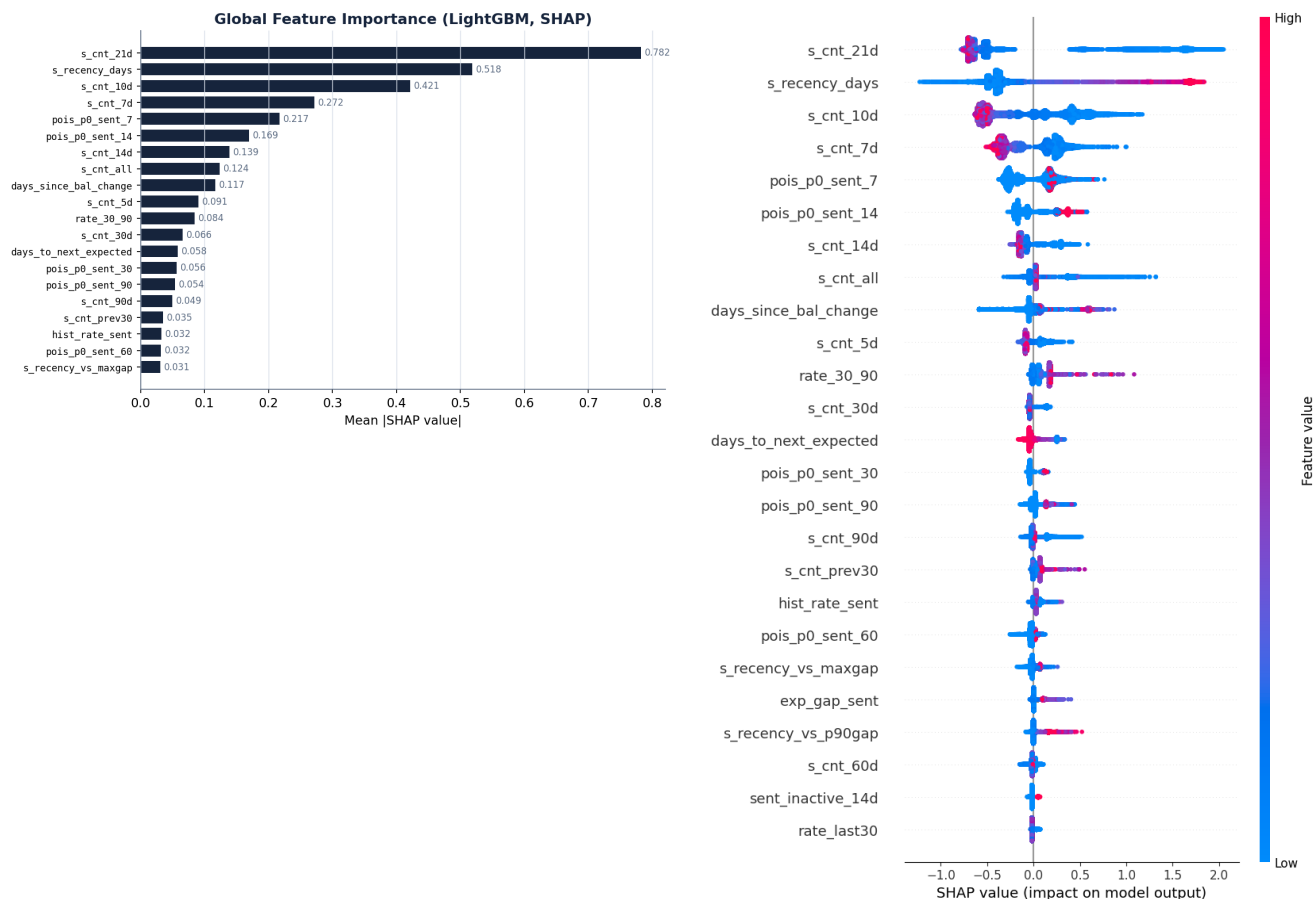


Figure 5 — Left: mean |SHAP| ranking (from the notebook's exported table). Right: the notebook's own SHAP beeswarm summary plot.

7. Leakage / artefact audit

On synthetic competition data a 0.985→0.999 jump usually comes from a generation artefact (label leaking through ID order, row order, or account metadata), not better modelling. Three direct probes were run. All sit at chance.

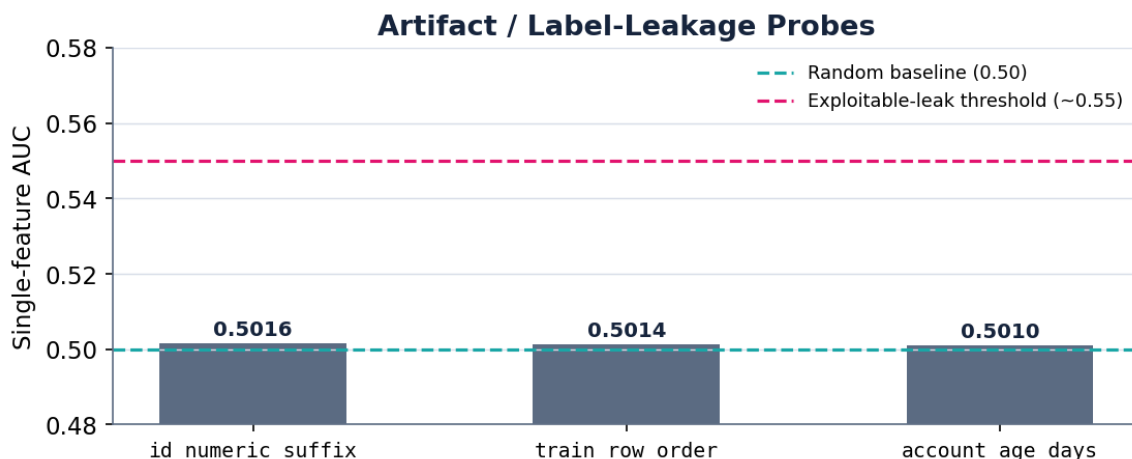


Figure 6 — Leakage probes (max AUC 0.5016). Nothing approaches the ~0.55 exploitable threshold, so the model's performance is bounded by genuine behavioural signal, not an artefact.

8. Honest assessment & limitations

The ensemble gain is marginal. The weight-optimised blend beats the best single LightGBM by +0.00002 OOF AUC. Differences at the fourth–fifth decimal are within seed/CV noise; the ensemble is kept because it does not hurt, not because it is decisively better.

Reaching 0.99 is not supported by this data. The strongest single feature plateaus at ~0.975 and the full model at ~0.985. The leakage audit is flat at 0.50, so there is no artefact to exploit. Closing the remaining ~0.014 gap would require signal that does not appear to be present; the residual is most consistent with irreducible noise given the

observation window.

Metric scope. This pipeline optimises and reports AUC-ROC, which is what the leaderboard scores. Any F1 figure would require a separate probability-threshold tuning step and would not change the AUC.

Reproducibility caveat. All figures above are copied from one executed run of the notebook. Tree seeds are fixed; the neural-net component and GPU kernels introduce small run-to-run variation (expected at the fourth decimal).